Vision-Guided Sample Handling for Laboratory Scanning Workflows Using a UR5e and an Eye-in-Hand RGB-D Camera

Abstract

Geological-sample inspection involves repeated loading, positioning, and return tasks that keep an operator at the instrument for hours. This report presents a vision-guided handling system for this task, assembled from off-the-shelf parts: a UR5e robot, a low-cost RGB-D camera, a parallel gripper, and 3D-printed fingertips, without custom fixtures or precision sensors. An eye-in-hand calibration relates camera observations to robot coordinates, and each sample is localised by global coarse detection followed by a centred RGB-D observation before grasping. Self-centring fingertips align the cylindrical samples mechanically with the gripping centre, reducing offsets that would otherwise be carried through transfer. The system moves samples from a tray to a workbench target and back to their original positions, and recovers from camera and robot-interface interruptions so that it can run unattended. The two-stage localisation retained all 15 tested positions and reduced the mean centre error from 2.820 to 1.243 mm compared with global connected-component estimation. The self-centring fingertips reduced the mean placement error from 1.887 to 0.759 mm, with all 100 placements within the 2 mm requirement. In a three-hour unattended run, the system completed 224 pick-and-return sequences and recovered from seven camera interruptions without a manual restart. These results provide a basis for integration with geological-sample inspection equipment.

Keywords: RGB-D vision; hand–eye calibration; centre estimation; self-centring gripper

1 Introduction

1.1 Application Background

Geological samples are often scanned one at a time. An operator places a sample in the scanner, waits for the scan to finish, takes the sample out, and puts in the next one. Some scanners use X-rays, some samples contain hazardous material, and after many repetitions mistakes become more likely. Robots are already used for routine laboratory work of this kind. Burger et al., for example, built a mobile robot that carried out chemistry experiments on its own [1].

Industrial robot cells load and unload parts all the time, but they are built around fixtures that hold every part in a known place and layouts that stay the same for years, and setting them up is costly. A research laboratory rarely works like that: there are fewer samples, budgets are smaller, and benches and procedures change from one project to the next. The loading itself is also quick. Most of the time goes into the scan, which lasts several minutes, so the operator has to stay near the instrument for the whole batch while spending only a short time on each sample.

1.2 Task Definition and Assumptions

In this project, the scanner is represented by a fixed target on a workbench. For every sample in the tray, the robot has to find it, pick it up, put it on the target, hold it there for the length of a simulated scan, and put it back in the same hole. The system has to meet two requirements. Each sample must be placed within 2 mm of the target centre, because its position in a real scanner decides which part of it is measured. In addition, the system must run unattended for the several hours of a batch, since replacing the operator's attendance is its purpose.

The size and shape of the samples, the hole pattern of the tray, the camera mount, the tool geometry, and the position of the scanner target are known and do not change. The table and tray are close to horizontal, and the tray is not moved during a pass. The robot does not know beforehand how many samples there are, which holes they are in, where the tray lies on the table, or where each sample sits inside its hole. The sample containers are translucent, so the depth camera often misses part of them.

Communication with a real scanner, identification of individual samples, obstacle avoidance, and work with people inside the robot's reach are not covered.

1.3 Related Work

Bohg et al. grouped grasp synthesis methods according to whether the objects are known, familiar, or unknown, and showed how prior knowledge of an object shapes the way grasps are generated and evaluated [2]. Since the sample geometry is known in this task, it can be used directly to estimate the sample centre and the grasp position.

The UR5e is widely used in manipulation research, which makes results obtained with it easier to reproduce and compare. Kang et al. combined it with a custom two-degree-of-freedom gripper for Soma-cube assembly [3], and Ali and Lee used it to validate regrasp planning for assembly tasks [4]. These studies show that a general-purpose arm often needs task-specific hardware or planning to meet the demands of a particular task. They address single assembly tasks, however, rather than the repeated, unattended handling required here (Section 1.2).

1.4 Contributions

The system is assembled from off-the-shelf parts rather than built as a dedicated industrial cell (Section 2.1). Only the fingertips and camera mount were made for the task, both by 3D printing, and there are no custom fixtures or precision sensors: instead of holding the tray at a known position, the system measures where the tray and samples are with a low-cost RGB-D camera. Even so, it ran unattended for three hours, completing 224 pick-and-return cycles and recovering automatically from every camera interruption (Section 6.2). The main contributions are:

- **Two-stage localisation.** A first view of the whole tray keeps samples whose depth data is incomplete, and a second, centred view of each sample gives the centre used for grasping (Section 4).

- **Self-centring fingertips.** As the gripper closes, the fingertips push the sample to the gripper centre. The vision system then only has to bring the sample within reach of the fingertips, so less accurate and possibly cheaper sensing could be enough. With these fingertips, all 100 test placements were within 2 mm (Section 7).

2 System Design

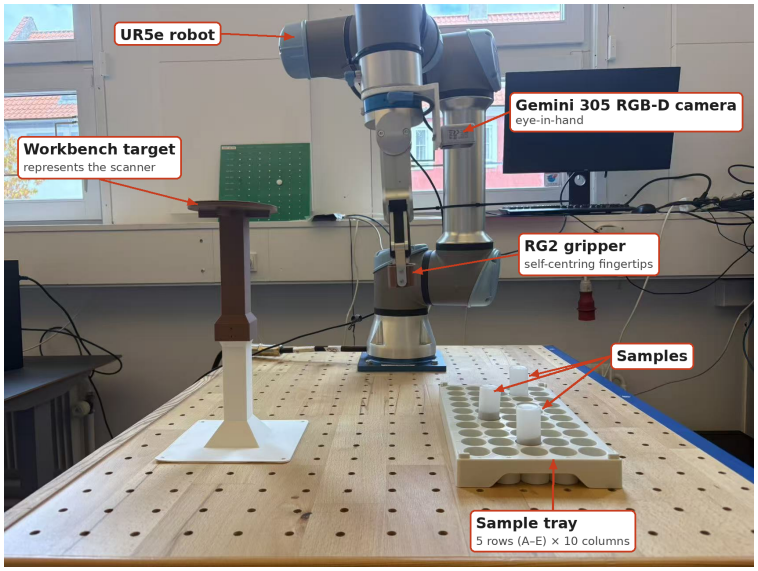
2.1 Hardware and Robot Interfaces

The system uses a UR5e collaborative robot, an OnRobot RG2 gripper, and an Orbbec Gemini 305 RGB-D camera, all controlled from a host computer running Python. The camera is connected over USB through the Orbbec SDK, and the robot controller over wired Ethernet through the interfaces described below.

With a payload of 5 kg and a reach of 850 mm, the UR5e can serve both the tray and the workbench without moving its base. Its pose repeatability of ± 0.03 mm is well below the 2 mm requirement, so placement errors come mainly from sensing, calibration, and the position of the sample in the gripper rather than from the arm itself. The RG2 has a stroke of up to 110 mm and a gripping force adjustable between 3 and 40 N, which is suitable for the 26.5 mm samples. Its fingertips can also be replaced with custom designs, such as those described in Section 7.1.

Some terms specific to Universal Robots are used in the rest of this report. The tool centre point (TCP) is the point on the end effector whose pose the controller reports and commands; it is defined as an offset from the robot flange and can be changed in the robot settings. The host computer talks to the controller through two interfaces. The Real-Time Data Exchange (RTDE) interface streams the robot state, including the TCP pose, and accepts motion commands. The Dashboard server accepts text commands that load, start, and stop programs and report their status. These programs are written in PolyScope, the graphical programming environment of the robot, and the RG2 gripper commands in this project are run in this way.

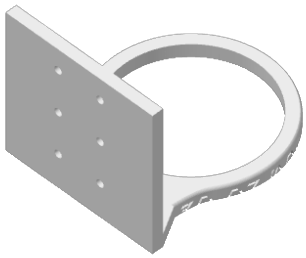
The Gemini 305 is attached to a 3D-printed mounting bracket beside the RG2 gripper, forming an eye-in-hand configuration. The rigid bracket keeps the camera fixed relative to the robot end effector, providing the constant geometric relationship required for hand–eye calibration. Figure 1 shows the complete system, the end-effector assembly, and the camera mounting bracket.



(a) Complete experimental system.



(b) Camera and gripper assembly.



(c) 3D-printed camera mount.

Figure 1: Hardware setup of the robotic sample-handling system.

2.2 System Workflow

Figure 2 summarises the handling sequence, and the hardware involved is labelled in Figure 1(a). Its steps are described in Sections 4 and 5.

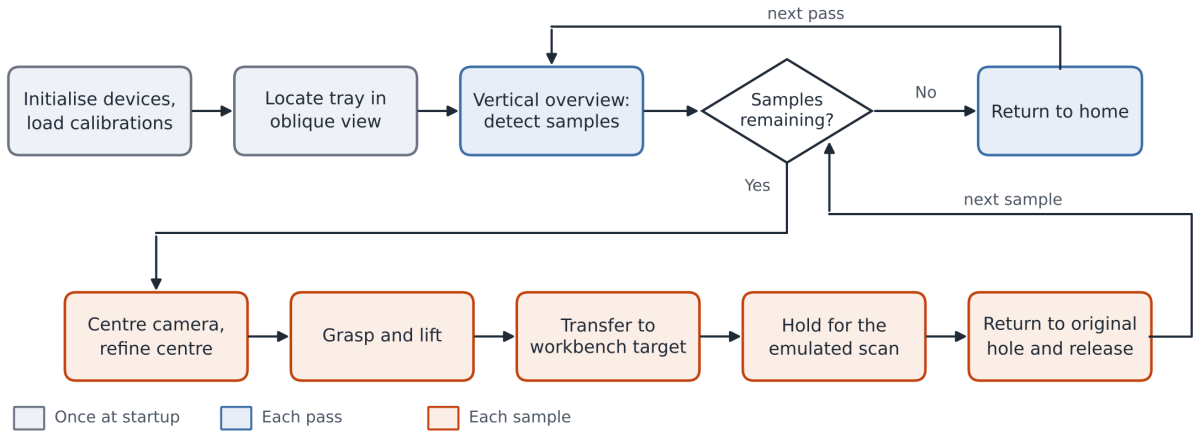


Figure 2: Sample-handling workflow. Colours indicate whether a step runs once at startup, once per pass, or once per sample.

2.3 Software Architecture

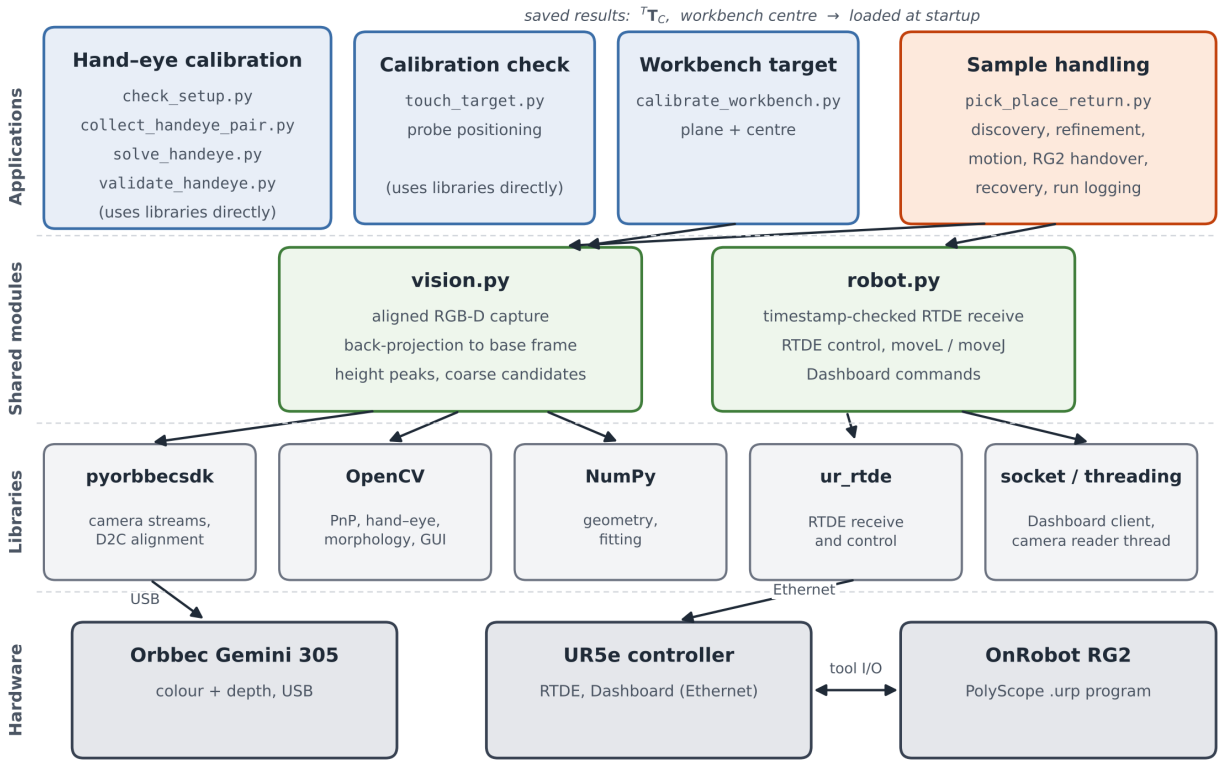


Figure 3: Software architecture of the sample-handling system.

The software is written in Python and organised as shown in Figure 3. Its core is `pick_place_return.py`, which runs the complete handling sequence: tray discovery, centred refinement, motion planning, the handover between motion control and the RG2 program, communication recovery, and run logging. It relies on two shared modules: `vision.py` for RGB-D capture and point extraction, and `robot.py` for robot state reading, motion commands, and Dashboard communication. These modules use the Orbbec SDK, OpenCV, NumPy, and `ur_rtde` to communicate with the camera and the robot controller. Separate scripts perform hand–eye calibration, the probe check, and workbench calibration before operation, and the main program loads their saved results at startup.

3 Coordinate Systems and Calibration

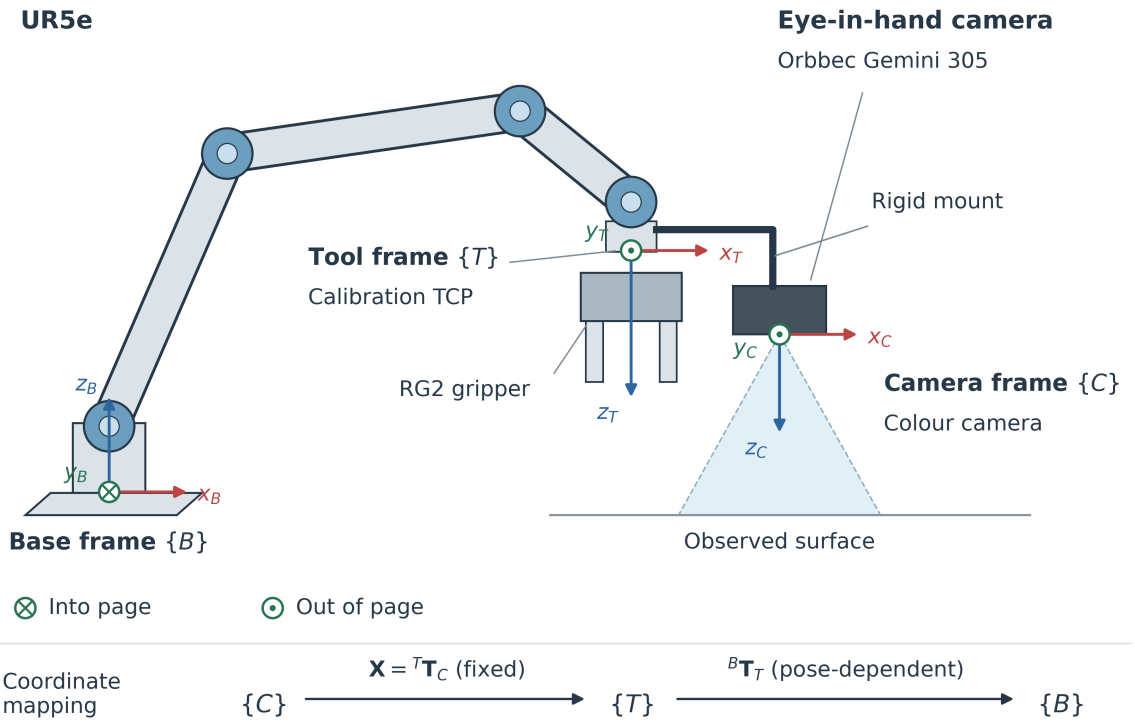


Figure 4: Coordinate frames of the eye-in-hand setup, geometry and frame orientations are schematic.

The robot moves to positions given in its base frame, whereas the sample positions are measured by the camera. Each camera measurement therefore has to be converted into the base frame before the robot can grasp the sample, and any error in this conversion carries over directly into the placement error. Figure 4 shows the frames involved, and Table 1 lists the entities of the setup in Section 2.1, how each is represented, and whether it is fixed. The camera frame $\{C\}$ has its origin at the optical centre of the colour camera, with its z-axis along the viewing direction. Of

the quantities in the table, only ${}^{T}\mathbf{T}_{C}$ cannot be read or measured directly: the bracket keeps it constant, but its exact value cannot be taken from the bracket design, so it is found by hand–eye calibration. A point seen by the camera is then mapped into $\{B\}$ through ${}^{B}\mathbf{T}_{T}\,{}^{T}\mathbf{T}_{C}$, where ${}^{a}\mathbf{T}_{b}$ maps coordinates from frame $\{b\}$ into frame $\{a\}$.

Table 1: Entities of the task and how their positions are obtained.

Entity	Representation	Fixed	How it is obtained
Robot base	$\{B\}$	Yes	Reference frame at the robot mount
Tool (flange)	${}^{B}\mathbf{T}_{T}$	No	From the pose reported by the controller
Camera	${}^{T}\mathbf{T}_{C}$	Yes	Hand–eye calibration
Workbench target	Point in $\{B\}$	Yes	Measured before operation
Tray	Rectangle in $\{B\}$	No	Detected in each pass
Samples	Points in $\{B\}$	No	Localised before each grasp

3.1 Hand–Eye Calibration

During calibration, the active TCP offset was set to zero, so that $\{T\}$ coincided with the robot flange. The procedure also uses a checkerboard frame $\{W\}$, which appears only in this subsection.

The colour-camera intrinsics and distortion coefficients were taken from the factory calibration of the Gemini 305, read through the Orbbec SDK for the colour-stream configuration used in this work. No further intrinsic calibration was carried out. The image configuration, the intrinsic matrix, and the distortion coefficients, in the order expected by OpenCV, are given in Appendix A.

The checkerboard has 6×8 inner corners with 25.5 mm squares. It was kept stationary while the robot viewed it from different poses, including rotations about non-parallel axes. In each image, the inner corners were detected with OpenCV. Because the pattern alone does not identify its origin, the operator selected the same physical corner in every image, and the corner grid was reordered so that the board's x-axis ran along the six-corner direction and its y-axis along the eight-corner direction.

The board pose relative to the camera was then estimated with `solvePnP` from the known corner coordinates and the camera parameters. The matching tool pose was read through RTDE, which gives the orientation as a rotation vector, and was converted to a rotation matrix with the Rodrigues formula. All translations are in metres. Images were taken only while the robot was stationary and were paired with the robot state received closest in time on the host, since the camera and robot are not hardware-synchronised.

For observation i, let

$$\mathbf{P}_i = {}^{B}\mathbf{T}_{T_i}, \qquad \mathbf{Q}_i = {}^{C_i}\mathbf{T}_{W}, \qquad \mathbf{X} = {}^{T}\mathbf{T}_{C}, \tag{1}$$

where $\{T_i\}$ and $\{C_i\}$ denote the tool and camera frames at the ith robot pose. Because the checkerboard does not move relative to the base and $\mathbf{X}$ is constant, ideal observations satisfy

$$\mathbf{P}_i\mathbf{X}\mathbf{Q}_i = {}^{B}\mathbf{T}_{W} = \mathbf{P}_j\mathbf{X}\mathbf{Q}_j. \tag{2}$$

Rearranging Equation (2) gives

$$\mathbf{A}_{ij}\mathbf{X} = \mathbf{X}\mathbf{B}_{ij}, \qquad \mathbf{A}_{ij} = \mathbf{P}_j^{-1}\mathbf{P}_i, \qquad \mathbf{B}_{ij} = \mathbf{Q}_j\mathbf{Q}_i^{-1}. \tag{3}$$

$\mathbf{X}$ was estimated from the measured pose pairs using the Park–Martin method implemented in OpenCV [5], which gave

$$^{T}\mathbf{T}_{C} = \begin{bmatrix} -0.999792 & -0.017278 & 0.010803 & 0.010021 \\ 0.017296 & -0.999849 & 0.001571 & 0.074890 \\ 0.010774 & 0.001757 & 0.999940 & 0.058478 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{4}$$

The translation is in metres. The values are rounded here, while the implementation keeps full precision.

With this result, the mapping introduced at the start of this section can be written for a point measured in image i as

$$^{B}\tilde{\mathbf{p}} = {}^{B}\mathbf{T}_{T_i}\,{}^{T}\mathbf{T}_{C}\,{}^{C_i}\tilde{\mathbf{p}}, \qquad {}^{C_i}\tilde{\mathbf{p}} = \begin{bmatrix} x_{C_i} & y_{C_i} & z_{C_i} & 1 \end{bmatrix}^{\mathsf{T}}, \tag{5}$$

where $\tilde{\mathbf{p}}$ denotes homogeneous coordinates and ${}^{B}\mathbf{T}_{T_i}$ is the tool pose at the time the image was taken. Depth points must be expressed in the calibrated colour-camera frame before this equation is applied; their extraction is described in Section 4.

During sample handling, the controller reports the pose of the active TCP, which may differ from the zero-offset TCP used for calibration. The tool pose is therefore recovered as

$$^{B}\mathbf{T}_{T} = {}^{B}\mathbf{T}_{\mathrm{TCP}}\left({}^{T}\mathbf{T}_{\mathrm{TCP}}\right)^{-1}, \tag{6}$$

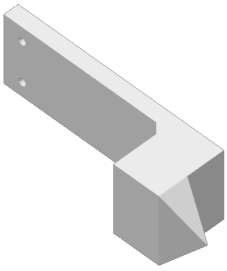
where ${}^{T}\mathbf{T}_{\mathrm{TCP}}$ is the configured TCP offset. In this way, the calibration remains valid when a different TCP is used. The transformation was checked with the probe-positioning test described in Section 3.2.

3.2 Calibration Verification

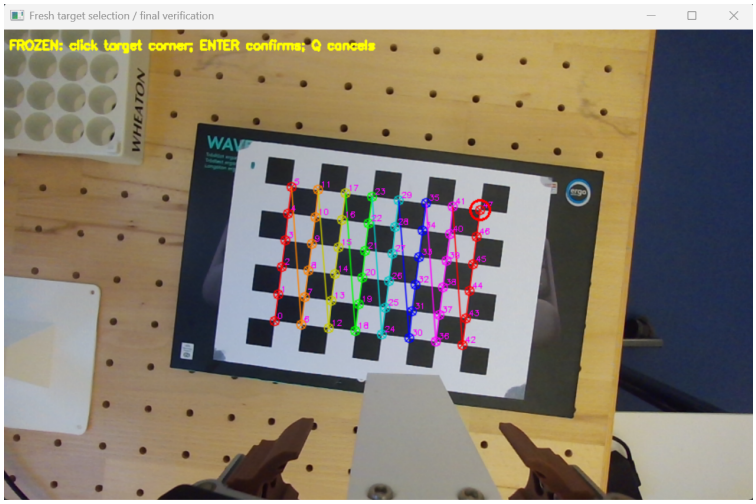
The calibrated hand–eye transformation was checked by moving a pointed probe to a selected checkerboard corner. The probe was designed around the RG2 gripper's dimensions and mounting details and produced by 3D printing (Figure 5a). The active TCP offset was kept at zero, and the nominal tip offset of 245.6 mm along the tool's positive Z-axis was included when calculating the target pose.

The robot first raised the probe and aligned it approximately with the normal of the detected checkerboard plane. Once it had stopped, a new image was captured and the operator selected an inner corner, shown in red in Figure 5b. The corner's three-dimensional position was calculated from the checkerboard geometry and estimated board pose, then transformed into the robot base frame. After confirming that the robot had not moved since image capture, the program moved the probe to the target at 10 mm/s without changing its orientation. The robot stayed at the final pose for visual inspection, with no automatic contact detection.

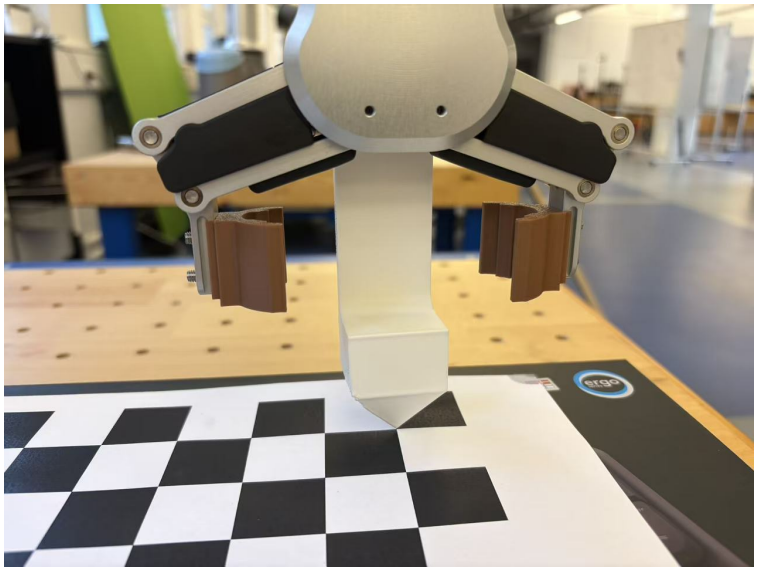
As shown in Figure 5c, the probe tip appears closely aligned with the selected corner. The result was considered acceptable for this positioning check, and work proceeded to vision-guided sample handling.



(a) Custom probe design.



(b) Target selection in the camera image.



(c) Probe positioned at the selected corner.

Figure 5: Probe-touching verification: the custom probe, the selected image target, and the physical positioning result. The red circle in (b) marks the selected checkerboard corner.

4 Visual Centre Estimation

4.1 Global Coarse Detection

Sample localisation proceeds in two stages, using the camera poses shown in Figure 6. The translucent samples can yield fragmented and uneven depth measurements, which may cause a sample to be rejected by strict boundary checks or shift its estimated centre. The global stage therefore accepts partial regions that remain compatible with the sample geometry and uses their centres only to reposition the camera. Centring then brings each sample close to the principal point, so that the viewing direction lies closer to the optical axis and to the cylinder axis. This reduces off-axis viewing but does not guarantee complete depth coverage. Section 4.3 compares the localisation errors of the two stages, and Table 7 lists the parameters.

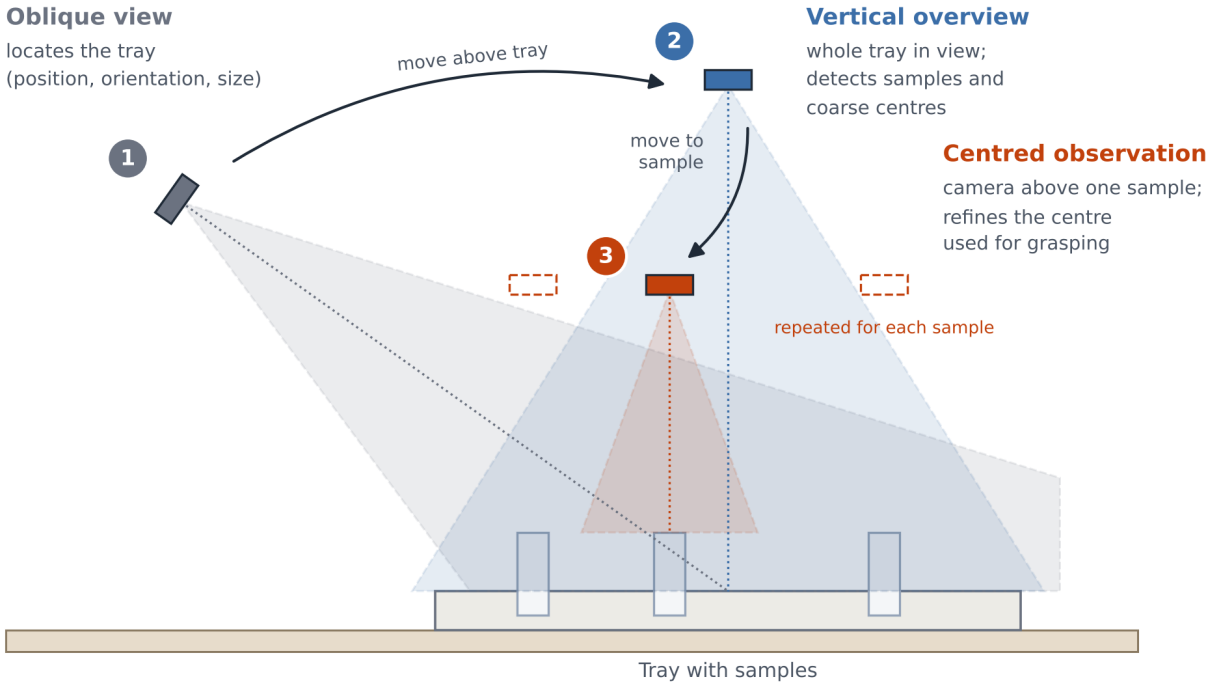


Figure 6: Camera poses used for localisation. Geometry is schematic.

The tray is located in the oblique view (Figure 7), assuming that the table and tray are approximately parallel to the base XY plane. Valid depth measurements are back-projected and transformed into the base frame using Equation (5). In the histogram of their heights, the table is taken as the low peak with the most supporting points, and the tray top as a sufficiently supported peak within the allowed height range above the table (Figure 7b). Points near the tray height form a mask, which is closed and divided into connected regions. Regions with too few points are discarded, a minimum-area rectangle is fitted in the base XY plane to each remaining region, and regions whose rectangles fall outside the tray-dimension limits are rejected (Figure 7c). The remaining regions are ranked by the product of rectangle area and supporting point count, and the

highest-scoring one gives the position, orientation, and size of the tray (Figure 7d).

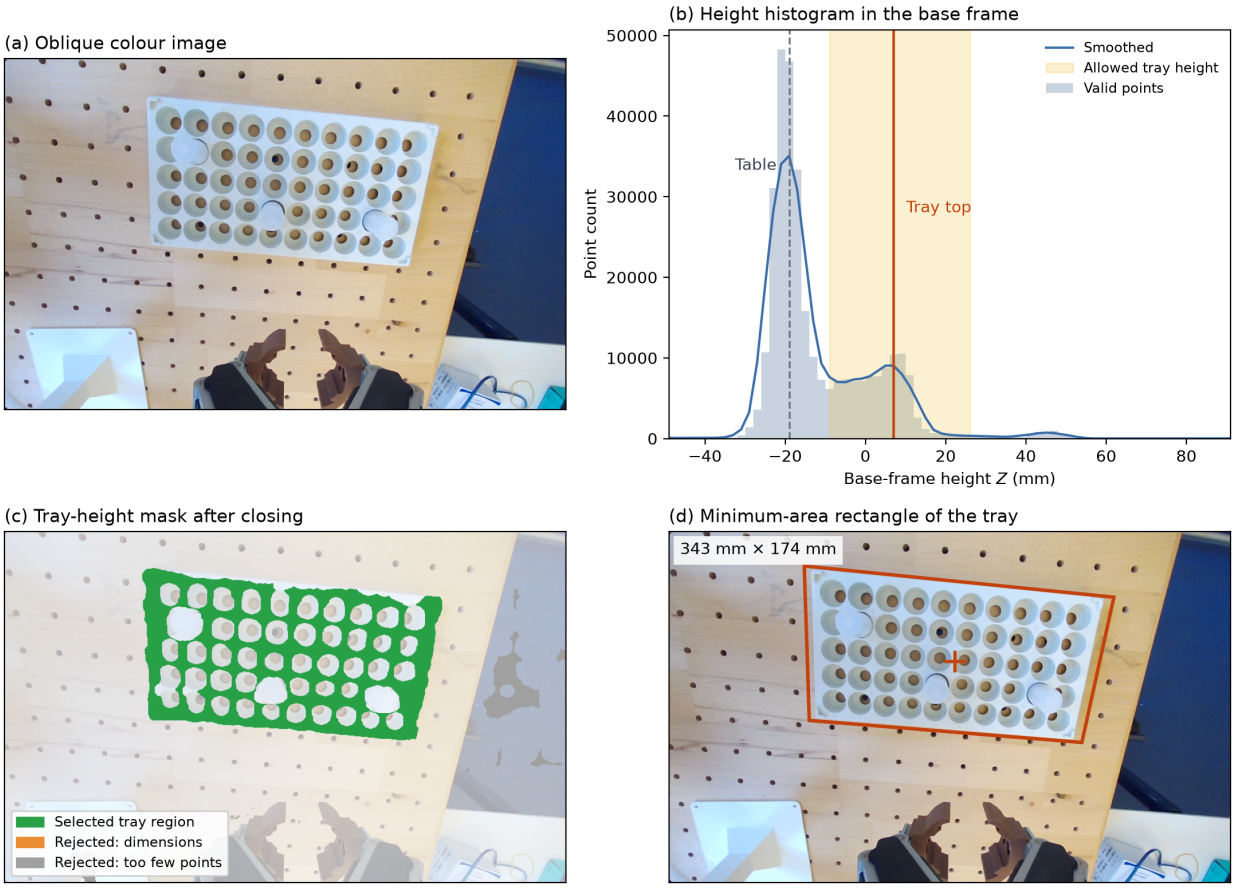


Figure 7: Tray detection in the oblique view. (a) Colour image. (b) Height histogram in the base frame. (c) Tray-height mask after closing. (d) Fitted rectangle projected into the image.

With the tray located, the samples are detected in a vertical overview of the tray. The robot moves the camera to a pose in which the horizontal image axis is aligned with the long axis of the tray and the whole tray is in view. Once the robot is stationary, several RGB-D frames are collected, and the temporal median of the valid depth values is computed at each pixel before back-projection; pixels with too few valid observations are left without depth. The median suppresses frame-to-frame fluctuations, but regions where depth is missing in every frame remain empty, so the subsequent steps must tolerate incomplete sample regions. Points within a height interval above the tray form the initial sample mask. Opening removes isolated regions, and after closing, the connected components are filtered by pixel area and physical extent.

A supplementary search with broader height and size limits recovers fragments missed by the initial segmentation, fragments that remain disconnected after closing are treated as separate candidates. Candidates from both searches are then checked loosely against the sample geometry, using the diameter, aspect ratio, and convex-hull circularity of the points in a band near the estimated top

height:

$$\gamma = \frac{4\pi A_{\text{hull}}}{P_{\text{hull}}^2}, \tag{7}$$

where A_{hull} and P_{hull} are the hull area and perimeter in the base XY plane. A candidate from the supplementary search is discarded if it lies within the separation threshold of a centre that has already been retained. Initial candidates are not compared with one another, so fragmentation can still produce duplicates.

For each retained candidate, $\mathcal{V}$ contains the pixels in its segmented component with finite three-dimensional coordinates. Each pixel j corresponds to ${}^{B}\mathbf{p}_j = [x_j, y_j, z_j]^{\mathsf{T}}$ through back-projection and transformation into the base frame. The coarse horizontal centre is

$$\mathbf{c}_{\text{coarse}} = \begin{bmatrix} \text{median}_{j\in\mathcal{V}}(x_j) \\ \text{median}_{j\in\mathcal{V}}(y_j) \end{bmatrix}. \tag{8}$$

The whole component is used because the top band alone may contain too few valid points. The median limits the influence of isolated outliers, but points on the side surface and an uneven distribution of valid points can still bias the estimate.

Finally, the tray rectangle is measured again in the vertical overview, so that boundary filtering uses the tray position observed together with the samples. Candidates outside the accepted interior of the tray are discarded, and the remaining ones are processed in order of increasing image distance from the principal point (Figure 8). Each of them then undergoes the centred refinement described in Section 4.2.

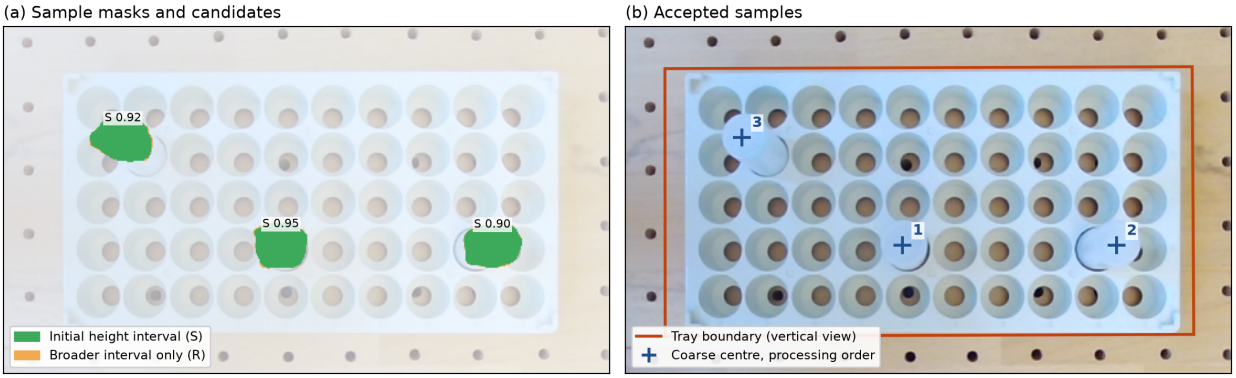


Figure 8: Sample detection in the vertical overview. (a) Sample masks; each label gives the search that produced the candidate (S: initial, R: supplementary) and its circularity γ. (b) Tray rectangle estimated from depth and accepted samples, numbered in processing order.

4.2 Centred Refinement

Using the coarse centre, the robot positions the camera so that its optical axis passes through the estimated top of the selected sample, while keeping the downward-looking orientation. After the robot stops, a new RGB-D sequence is acquired and processed with the temporal median described in Section 4.1. Figure 9 follows the refinement of one sample.

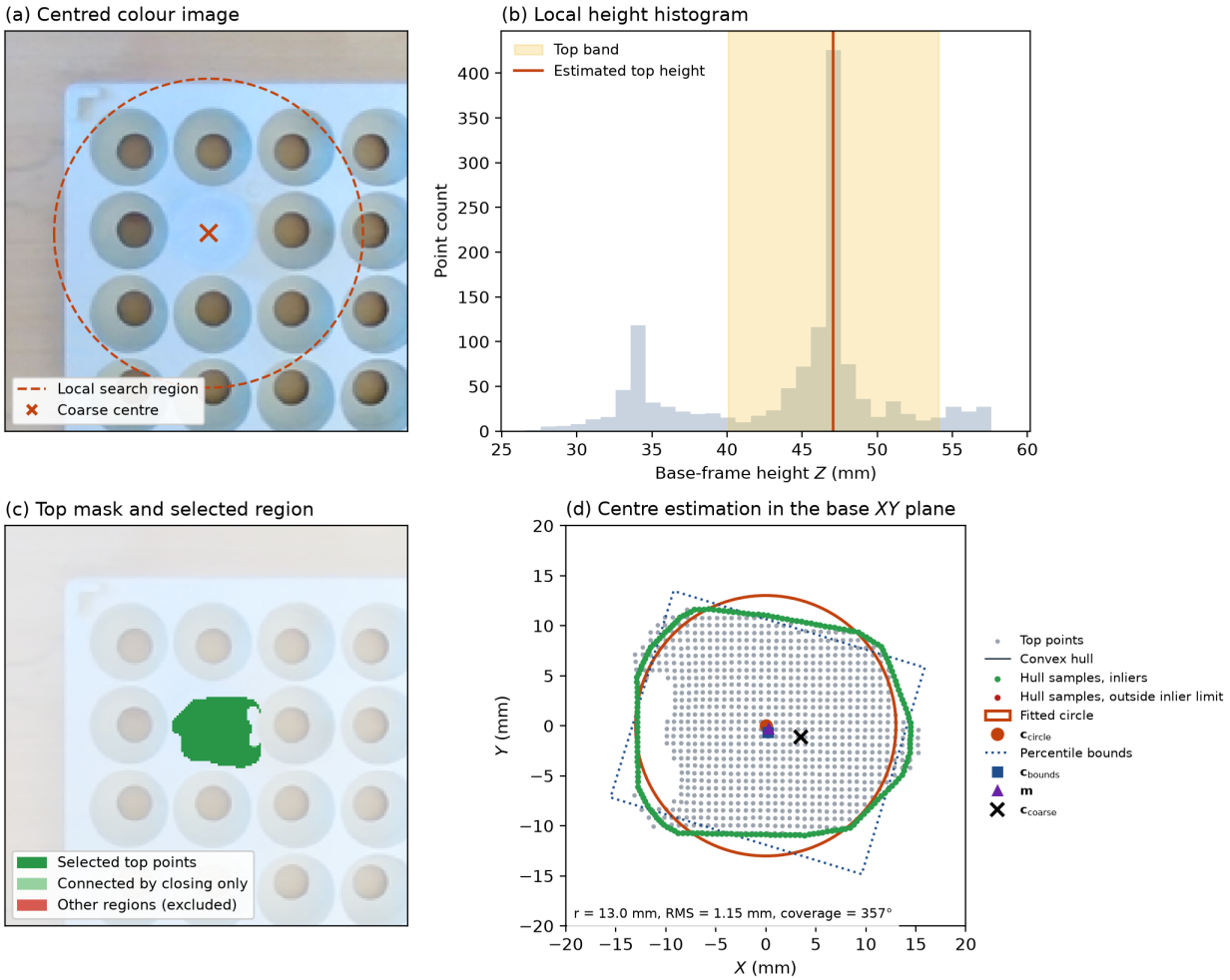


Figure 9: Centred refinement of one sample. (a) Centred image and local search region. (b) Local height histogram and top band. (c) Selected top region. (d) Top points, convex hull, fitted circle, percentile bounds, and the centre estimates in the base XY plane, plotted relative to the accepted centre.

Valid depth measurements are transformed into the base frame. A local search around the coarse centre excludes neighbouring samples, and height limits remove points near the tray and above the expected sample region (Figure 9a). The most populated bin of the local height histogram gives an estimate of the top height, and the points within a band around it form the top mask (Figure 9b).

After morphological closing, the components of the mask are ranked by their distance from the coarse centre, their image distance from the principal point, and their valid point count. The lowest-scoring component becomes the seed, and nearby components are added as possible fragments of the same top surface. Only measured points from the original top mask are kept, because closing establishes connectivity without creating new three-dimensional measurements (Figure 9c).

The selected points are projected onto the base XY plane, and points far from their coordinate-wise median are trimmed. A convex hull is built from the remaining points and sampled at approximately

uniform intervals along its boundary, so that densely sampled parts of the outline do not dominate the fit. An initial circle is obtained with an algebraic least-squares formulation of the type described by Kåsa [6]. For each sampled boundary point $\mathbf{q}_k$, the radial residual is

$$e_k = \|\mathbf{q}_k - \mathbf{c}\|_2 - r, \tag{9}$$

where $\mathbf{c}$ and r are the circle centre and radius. Refinement uses the Huber loss

$$\rho_\tau(e) = \begin{cases} \frac{1}{2}e^2, & |e| \le \tau, \\ \tau|e| - \frac{1}{2}\tau^2, & |e| > \tau. \end{cases} \tag{10}$$

For a given scale τ_{H}, the fitting objective is

$$\mathcal{L}(\mathbf{c}, r) = \sum_k \rho_{\tau_{\mathrm{H}}}(e_k). \tag{11}$$

Starting from the algebraic fit, each iteration linearises the radial residuals and solves a weighted Gauss–Newton step for the centre and radius. The weights are $w_k = 1$ when $|e_k| \le \tau_{\mathrm{H}}$ and $w_k = \tau_{\mathrm{H}}/|e_k|$ otherwise, and the scale is recalculated from the median absolute deviation of the residuals at each iteration.

Where part of the top region is missing, the convex hull bridges the gap with a straight chord (Figure 9d). Samples on this chord also enter the fit and are down-weighted only when their residuals exceed the Huber threshold, so their exclusion is not guaranteed. The fitted circle is therefore accepted only if its radius, inlier RMS residual, and angular coverage lie within their limits. Angular coverage is calculated by sorting the inlier boundary samples by their angle around the fitted centre and subtracting the largest angular gap from a full revolution. Because these samples come from the convex hull, this measure does not show how much of the physical outline was actually observed.

A second estimate uses the spatial extent of the trimmed top points. Let $\mathbf{s}_j \in \mathbb{R}^2$ denote these points and $\mathbf{m}$ their coordinate-wise median. The orthogonal matrix $\mathbf{E}$ contains the principal directions obtained from their covariance matrix, and each point is expressed in these directions as

$$\mathbf{a}_j = \mathbf{E}^{\mathsf{T}} (\mathbf{s}_j - \mathbf{m}) . \tag{12}$$

The vectors $\boldsymbol{\ell}$ and $\mathbf{u}$ contain the component-wise 2nd and 98th percentiles of $\mathbf{a}_j$, and their midpoint gives the bounds centre

$$\mathbf{c}_{\mathrm{bounds}} = \mathbf{m} + \mathbf{E}\frac{\boldsymbol{\ell} + \mathbf{u}}{2}. \tag{13}$$

Using percentiles rather than extreme values limits the influence of outlying points on the estimated extent.

If the circle passes its checks, its centre is retained only if it agrees with $\mathbf{c}_{\mathrm{bounds}}$ within the allowed limit. Otherwise, $\mathbf{c}_{\mathrm{bounds}}$ is used only if it agrees with $\mathbf{m}$. Excessive disagreement in either case rejects the estimate. Since $\mathbf{c}_{\mathrm{bounds}}$ and $\mathbf{m}$ are computed from the same points, their agreement reveals large inconsistencies but does not guarantee an unbiased result. The selected centre must also lie within the permitted distance of the coarse position. A failed refinement triggers one additional capture at the same pose, and a second failure stops the sequence before grasping. The accepted horizontal centre replaces the coarse estimate for grasping. In the example of Figure 9, the refined centre lies 3.7 mm from the coarse centre, and the circle and bounds estimates agree within 0.7 mm. The grasp and transfer motions built on this centre are described in Section 5.

4.3 Comparison of Centre-Estimation Methods

The two-stage procedure described above was evaluated by comparing global connected-component estimation, global single-frame circle fitting, and centred refinement. The comparison considers detection coverage and centre-estimation error.

Reference positions were obtained using a custom 3D-printed locating tool attached to the robot end effector (Figure 12b). Its cylindrical section was inserted into each selected tray hole to align the locating axis with the hole centre. The reference XY coordinates were then calculated from the recorded robot pose and the known relationship between the locating axis and the active TCP. These measurements provided a mechanical reference independent of visual centre estimation. Because the sample diameter was smaller than the hole diameter, the sample was manually placed as close as possible to the hole centre before each measurement. Any remaining offset between the two centres was therefore included in the measured error.

Fifteen holes spanning the middle and side regions of the tray were selected, as shown in Figure 12a. The tray has five rows labelled A–E and ten columns numbered 1–10 along its long side. For position i and repetition k, the error relative to the measured hole centre is

$$e_{ik} = \left\| \hat{\mathbf{c}}_{ik} - \mathbf{c}_i^{\mathrm{ref}} \right\|_2, \tag{14}$$

where $\hat{\mathbf{c}}_{ik}$ is the estimated sample centre in the robot base XY plane and $\mathbf{c}_i^{\mathrm{ref}}$ is the mechanically measured hole centre. Errors were averaged over repeated measurements at each position. The reported mean, median, RMSE, and maximum are calculated across these position-level means, with RMSE defined as their root mean square. Detection coverage is reported as the number of tested positions yielding valid estimates.

Global connected-component estimation and centred refinement follow Sections 4.1 and 4.2. The single-frame baseline fits a circle using one depth frame captured at the global overview pose, without camera repositioning or temporal depth aggregation. A position is counted as valid only when the circle fit passes the baseline's acceptance checks. Rejected fits are recorded as missed positions and excluded from its error statistics.

Table 2: Detection coverage and centre errors of the three methods.

		Error (mm)			
Method	Valid positions	Mean	Median	RMSE	Maximum
Global connected-component centre	15/15	2.820	2.867	2.952	4.069
Global single-frame circle fitting	11/15	2.085	2.126	2.275	3.551
Centred refinement	15/15	1.243	1.369	1.289	1.747

Connected-component detection identified all 15 positions, while single-frame circle fitting returned valid estimates at only 11 (Table 2). Coarse detection based on connected components therefore retained samples whose global-view depth regions did not yield an accepted circle fit. Centred refinement also retained all 15 positions and reduced the mean error by 55.9% compared with

connected-component estimation. Its mean error was 40.4% lower than that of the single-frame baseline, although the two means describe different sets of valid positions, and its median, RMSE, and maximum were also the lowest of the three methods.

The regional comparison in Appendix E shows that refinement also reduced the difference between the middle and side regions. These results support the two-stage approach: coarse detection preserved coverage across the tested positions, while the centred observation brought the estimated centres closer to the mechanical references.

5 Sample Handling and Motion Control

5.1 Motion Target Definition

The motion sequence combines the refined sample position from Section 4.2 with a workbench target measured before operation. A separate RGB-D procedure identifies the elevated workbench surface, estimates its plane, and calculates the centroid of the enclosed region in the base XY plane. Colour-image edges refine the boundary only when they provide enough perimeter support and the resulting centre remains within the allowed distance from the depth-based estimate. The centre is then saved as the fixed workbench target.

The refined sample centre determines the horizontal grasp coordinates, while the TCP height is obtained by adding a grasp offset to the estimated seating level beneath the sample. This level is identified from a lower height peak below the tray's upper surface. If no suitable peak is found, it is estimated by subtracting the specified recess depth from the upper-surface height.

The workbench TCP height uses the same grasp offset, added to the workbench surface height. If the sample retains its position in the gripper, its bottom therefore lies nominally at the surface while the fingers remain closed. The target includes no additional hovering gap, and the dwell uses position control without force regulation. Actual contact consequently depends on the surface-height estimates and the sample's position within the gripper.

Before grasping, the system calculates the approach, grasp, intermediate, and workbench poses and checks them for reachability and compliance with the controller's safety limits. The intermediate-position search begins at the midpoint between the sample and workbench. At each candidate position, it tests the permitted vertical-axis orientations and workbench approach heights, accepting the first complete pose set that passes. Cylindrical symmetry allows this orientation to vary during selection, after which it remains fixed throughout transfer and return. If no pose set passes, execution stops before grasping. These checks cover destination poses rather than every point along the linear movements. The route therefore assumes a clear transfer corridor in the fixed experimental layout and includes no obstacle model of the arm or end-effector assembly.

5.2 Picking, Transfer, and Return

With the gripper open, the robot moves above the refined sample centre and descends vertically to the grasp pose. After the RG2 closes, the robot lifts the sample to the tray-side transfer height and moves horizontally to the intermediate position. The height changes here because the tray and

workbench require different transfer heights. From this position, the robot continues horizontally above the workbench centre and descends to the target pose.

The waypoint heights account for the estimated sample top, workbench surface, and required clearances. All movements between waypoints use linear Cartesian commands, with lower speed and acceleration during vertical approaches and withdrawals. The motion settings and workbench dwell time are given in Table 8.

Each motion or gripper action finishes before the next begins. The RG2 runs through a separate PolyScope program, which requires the RTDE control script to stop before the gripper command starts. The system checks the output-register acknowledgement together with the Dashboard program state, then recreates the control interface after the gripper program finishes. These checks confirm that the command has completed, but do not independently confirm that a sample is held.

At the workbench, the gripper stays closed during the dwell period to simulate the wait for sample inspection. The robot then lifts the sample and follows the intermediate waypoints in reverse order. It descends to the saved grasp pose in the tray, opens the gripper, and withdraws vertically once opening is complete. This return movement assumes that the sample has not slipped significantly within the gripper and that its original position remains available. The tray must remain stationary during transfer and return, as the sample is not visually located again while held.

The same sequence is repeated for the remaining samples, with a new centred observation before each grasp. After all samples have been handled, the robot returns to the home pose, defined as the joint configuration recorded before the initial camera-alignment movement. It then moves back to the overview pose and repeats global coarse detection to obtain the sample list for the next pass.

6 Communication Recovery and Continuous Operation

The task definition in Section 1.2 requires the system to run unattended for several hours, because its purpose is to replace the operator who would otherwise attend the scanner throughout a batch. Over such periods, the camera stream or the connection to the robot controller can be interrupted, and a single interruption should not end the run or require someone to restart it. This section describes how these interruptions are detected and recovered from, and how the system stops when recovery is not possible. The resulting behaviour is evaluated in the three-hour continuous run in Section 6.2.

6.1 Communication Recovery

The system includes recovery procedures for RGB-D acquisition and the receive and control interfaces of the ur_rtde library [7]. Attempts are limited and separated by delays, with the main settings listed in Table 9.

The camera uses a continuous reading thread to prevent old frames from accumulating while no capture is requested. Reader errors, acquisition timeouts, or the absence of a new valid aligned frame within the centred-acquisition progress timeout trigger a pipeline restart. The existing pipeline and reading thread are closed, and a new camera session is opened after a delay. Initial frames are

discarded before the interrupted acquisition is repeated. The centred acquisition has a larger retry allowance because its observation is used to calculate the grasp position.

Acquisition validity is checked separately from stream health. Before capture, TCP linear and angular velocities must remain below their limits during a settling interval. Robot poses and active TCP settings are also compared before and after acquisition. Robot movement or a changed TCP setting invalidates the observation rather than triggering a camera-stream restart. The repeated observation after a failed centre estimate, described in Section 4.2, is a separate visual-quality retry.

Before returning robot-state data, the receive manager checks that the controller timestamp has advanced. This prevents cached values from being accepted after communication has stopped. Automatic reconnection is enabled for stationary pose snapshots and RG2 acknowledgement reads, when no arm movement is being commanded. If one of these reads fails or its timestamp stops updating, the existing interface is disconnected and replaced. The replacement must receive advancing timestamps before supplying data.

Establishing the RTDE control interface is retried if interface creation fails or the new connection is unavailable. A longer retry schedule is used after an RG2 program to allow for the handover back to the motion-control script described in Section 5.2. These retries cover connection setup rather than interruptions during an executing movement. Interrupted motion commands are not automatically replayed.

The sequence ends when communication, observation, motion, or gripper-command failures remain unresolved after the applicable recovery attempts. The system attempts to stop motion without automatically releasing the gripper or returning to the home pose. There is no separate application-level timeout for a motion call that remains blocked. The current cycle, sample, execution stage, and error information are recorded, together with messages identifying connection attempts and successful interface reconstructions.

6.2 Continuous Operation and Cycle Timing

During the three-hour test, the system completed 224 pick-and-return sequences, with a mean elapsed time of 48.2 s per sample. The tray held three samples, so each pass described in Section 5.2 handled three samples. The session was recorded on video, and a 60× time-lapse version is available on GitHub.[1] The source code is listed in Appendix G.

Stage timings were measured across repeated trials and are summarised in Table 3. Because they were measured separately, they do not form an exact breakdown of the three-hour run, and the overall mean of 48.2 s per sample also includes shared cycle overhead and pauses for communication recovery.

This cycle time may appear slow for an industrial arm, but it follows from deliberate choices rather than from the limits of the hardware. Horizontal transfers run at 0.30 m/s and vertical approaches at 0.08 m/s (Table 8), well below the typical TCP speed of 1 m/s of the UR5e, so that the held sample is not disturbed and the robot moves predictably. The sequence is also strictly sequential: each capture waits for the robot to come to rest and combines several frames, each sample requires

[1] Extended-operation video on GitHub.

an additional move to the centred view, and each gripper command involves switching between the motion-control script and the gripper program. For the intended application, throughput is set by the scanner rather than by the robot, since a scan lasts several minutes; the requirement that matters is unattended operation over a full batch. Ways to shorten the cycle are outlined in Section 8.2.

With the robot connected over wired Ethernet, RTDE receive and control disconnections were rarely observed during repeated experiments. Seven camera-stream interruptions occurred during the three-hour run, and all seven were recovered automatically without a manual restart. The recovery times in Table 3 come from separate interruption tests rather than from these seven events.

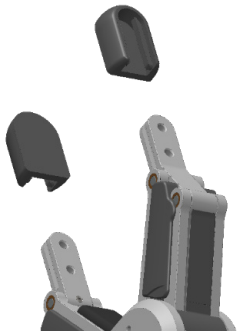
The run demonstrates sustained sample handling despite temporary camera interruptions. Its three-hour duration reflects the recorded test period rather than a programmed operating limit; the handling sequence continues until it is interrupted or stopped by an unresolved error.

Table 3: Mean execution times. Group headings state how often each stage occurs.

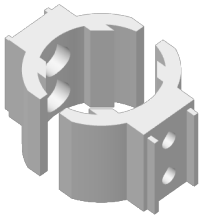
Stage	Time (s)
Once at startup	
Setup and warm-up	18.0
Once per pass	
Global tray and sample discovery	10.5
Per sample	
Centred localisation	5.7
Grasping	10.3
Transfer to the workbench	8.6
Workbench dwell	2.0
Return to the tray	10.7
Overall	
Mean elapsed time per sample, incl. recovery pauses	48.2
Camera recovery (separate tests)	
After one stream interruption	15.0
After two consecutive interruptions	34.0

7 Self-Centring Fingertips

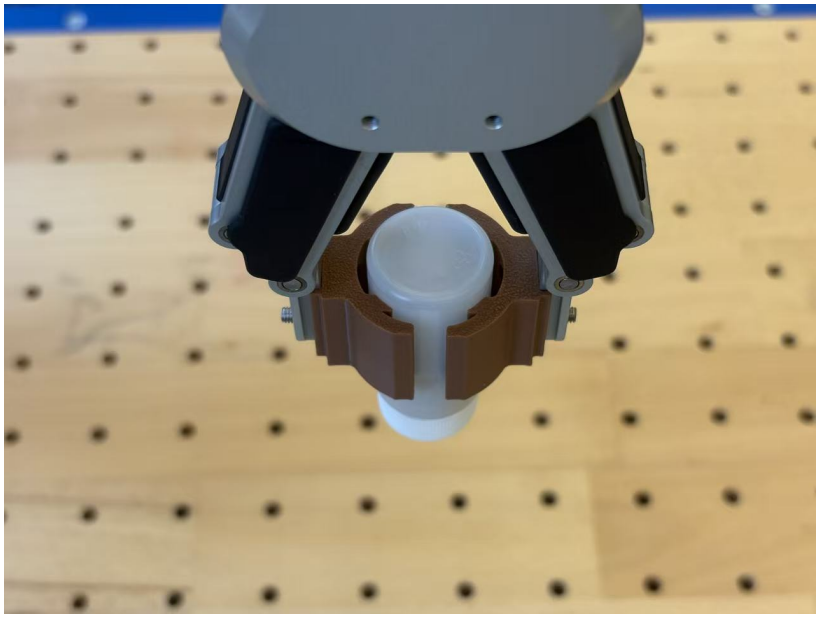
7.1 Motivation and Design



(a) Original RG2 fingertips.



(b) Self-centring fingertip design.



(c) Printed fingertips holding a sample.

Figure 10: Original and self-centring fingertips. The original fingertip illustration is reproduced from the OnRobot RG2 datasheet.

Two sources of error remain after visual localisation. The refined centre still contains a residual error, and together with hand–eye calibration, TCP definition, and robot positioning errors, it can produce a placement offset larger than the localisation error alone. In addition, the sample diameter of about 26.5 mm is smaller than the 30 mm tray hole, leaving a nominal radial clearance of 1.75 mm. The sample can therefore sit off-centre in its hole and move when the closing fingertips touch it, so that it ends up held off-centre. If this offset is kept during transfer, it adds to the final placement error and to the variation between trials.

Both errors appear as a lateral offset of the sample relative to the gripper. New fingertips were therefore designed to correct this offset during closing, instead of relying on more accurate sensing. The idea draws on work showing that contact mechanics can take over tasks otherwise left to sensing and actuation. Rodriguez and Mason showed that finger shapes can make the grasp of a disk invariant to its size and position, with straight-line and logarithmic-spiral profiles as solutions [8], and extended this approach to effector design for general one-degree-of-freedom mechanisms [9]. Chavan-Dafle et al. demonstrated that a simple gripper can reposition an object in the hand by exploiting external contacts and gravity instead of additional actuation [10].

The fingertips developed here apply this principle with straight locating faces, the simplest of the profiles above (Figure 10b). Each fingertip carries two axial ribs whose working faces form a 90° V, so the two opposing fingertips provide four symmetric locating faces that touch the cylinder at four places around its cross-section. The surrounding curved walls stay clear of the sample and leave room for it to enter and shift, so its final lateral position is set by the four faces alone. The ribs extend over an axial length of 26 mm to support the sample and limit tilting. Figure 10c shows the printed fingertips holding a sample.

Figure 11 illustrates how this geometry centres the sample. Flat jaws constrain the sample only along the closing direction, so an offset perpendicular to it remains after closing (Figure 11a). Each locating face is instead inclined at 45° to the closing direction. When the sample is off-centre, it first meets the faces on one side, and the contact forces then have a lateral component that pushes it towards the centre line (Figure 11b). Once the sample rests against all four faces, its axis coincides with the gripping axis (Figure 11c). This holds when the fingers close symmetrically and the sample can slide sideways; the actual effect also depends on friction and on printing and assembly tolerances. The design centres the sample laterally but does not locate it along its axis.

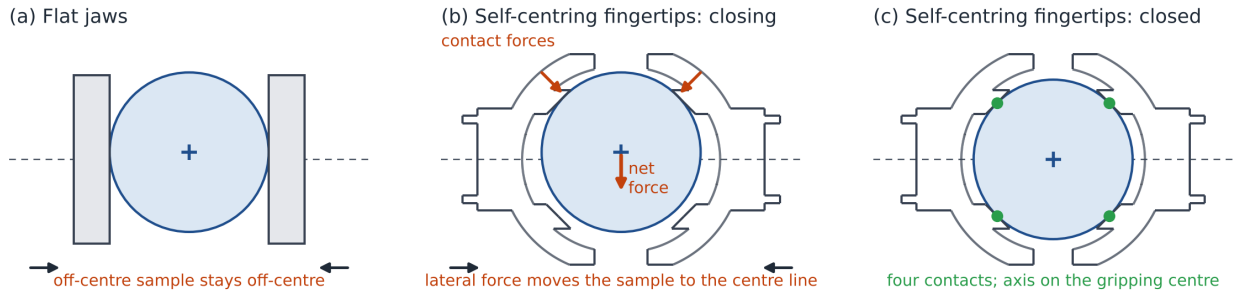


Figure 11: Centring principle, shown on a cross-section of the printed fingertips. (a) Flat jaws leave a lateral offset. (b) The inclined faces push an off-centre sample towards the centre line. (c) The sample rests against all four faces.

7.2 Placement Accuracy

The two fingertip designs were compared over 100 placements each. In every trial, the sample was located by global coarse detection and centred refinement, grasped, and transferred to the saved workbench centre. Unlike the continuous sequence in Section 5.2, the gripper then opened and withdrew, leaving the sample on the workbench for measurement. The measured error therefore also includes any displacement during release.

The measurements were made on a 3D-printed workbench with a symmetric elliptical top and scales along its two symmetry axes (Appendix F). After each placement, the gaps between the sample and the opposite edges of the workbench were read manually from these scales: the left and right gaps along a common line parallel to one symmetry axis, and the upper and lower gaps along a common line parallel to the other. With L, R, U, and D denoting the left, right, upper, and lower gaps, the horizontal offset of the sample centre is

$$e_{\text{place}} = \frac{1}{2}\sqrt{(R-L)^2 + (U-D)^2}, \tag{15}$$

which is zero when opposite gaps are equal. Both designs were measured in the same way, and the results are limited by the precision of the manual readings.

Table 4: Placement errors over 100 trials with each fingertip design.

Metric	Original fingertips	Self-centring fingertips
Mean error	1.887 mm	0.759 mm
Median error	1.762 mm	0.707 mm
Standard deviation	0.709 mm	0.383 mm
Minimum–maximum	0.640–3.202 mm	0.100–1.900 mm
Error ≤ 1 mm	8/100 (8%)	71/100 (71%)
Error > 2 mm	40/100 (40%)	0/100 (0%)

With the self-centring fingertips, the mean placement error fell by 59.8% and the standard deviation by 46.0% (Table 4). All 100 placements met the 2 mm requirement defined in Section 1.2, whereas 40 of those made with the original fingertips did not. The centring geometry thus complements visual localisation by removing much of the offset that remains after grasping, transfer, and release.

8 Conclusions and Future Work

8.1 Conclusions

This project demonstrates repeated picking and return of cylindrical samples for geological inspection. Global coarse detection retained all tested positions, while centred refinement improved centre estimation. The self-centring fingertips reduced the placement errors remaining after visual localisation, grasping, transfer, and release, giving more accurate and consistent placement than the original fingertips.

During the three-hour test, the system continued operating through seven camera interruptions without a manual restart. This demonstrates sustained sample handling under the tested conditions. The resulting workflow provides a basis for integration with inspection equipment, where instrument feedback would replace the current workbench dwell.

8.2 Improvements and Future Work

The experiments showed that the system can carry out repeated sample handling, but grasp verification, interruption recovery, gripper response time, and throughput could still be improved. These changes would also support its integration with inspection equipment.

Grasp verification could be added before sample transfer. The current gripper acknowledgement confirms that a command has completed, but does not independently confirm that a sample is held. Gripper-width or grip-detection feedback could identify empty grasps before the robot leaves the tray.

An interruption-recovery procedure could use the recorded execution stage, current robot state, and grasp feedback to determine whether a sample remains held. If so, it could return the sample to its saved tray position once communication is restored and the return path is confirmed clear. This would help keep the sample secure after an interrupted run.

Gripper control could also be made more efficient. Opening and closing currently rely on a separate `.urp` program stored on the UR5e controller. Switching between this program and the RTDE control script introduces delays into the handling cycle. A supported control interface that avoids repeated program switching could reduce this overhead while retaining command acknowledgement.

Throughput could be increased further if required, for example with a faster instrument. Speeds and accelerations could be raised within a risk assessment, and consecutive waypoints could be blended rather than stopping at each one. Using fewer frames per capture would shorten each observation, and where the self-centring fingertips tolerate the coarse localisation error, the centred observation might be omitted. With a real scanner, the next sample could also be localised while the current one is being scanned, so that localisation no longer adds to the cycle time.

The next application step is integration with an actual geological-sample inspection system. The current workbench dwell represents the waiting stage of inspection. Instrument integration would require a communication interface to confirm that the sample is ready, start the measurement, and wait for completion before returning the sample to the tray.

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A Factory Camera Calibration Parameters

A.1 Image Configuration

Table 5: Colour-stream configuration used during hand–eye calibration.

Parameter	Setting
Camera	Orbbec Gemini 305
Sensor	Colour
Image resolution	848×530 pixels
Frame rate	15 fps
Pixel format	YUYV
Distortion model	Brown–Conrady K6

A.2 Intrinsic Matrix

The intrinsic matrix used in the implementation was

$$\mathbf{K} = \begin{bmatrix} 409.382324 & 0 & 421.535278 \\ 0 & 409.362946 & 265.943817 \\ 0 & 0 & 1 \end{bmatrix}. \tag{16}$$

The focal lengths are $f_x = 409.382324$ and $f_y = 409.362946$, and the principal point is $(c_x, c_y) = (421.535278, 265.943817)$. All four quantities are expressed in pixels.

A.3 Distortion Coefficients

The distortion coefficients were supplied to OpenCV in the order

$$\mathbf{d} = \begin{bmatrix} k_1 & k_2 & p_1 & p_2 & k_3 & k_4 & k_5 & k_6 \end{bmatrix}. \tag{17}$$

The coefficients $k_1, \ldots, k_6$ describe radial distortion, while p_1 and p_2 describe tangential distortion. Their dimensionless values are listed in Table 6.

Table 6: Factory distortion coefficients used in the implementation.

Coefficient	Value
k_1	-1.245939135551
k_2	0.817450463772
p_1	0.000206331912
p_2	0.000061016010
k_3	-0.182974949479
k_4	-1.228488683701
k_5	0.794195711613
k_6	-0.173994883895

The values presented here are rounded for readability. The implementation retains the full numerical precision of the exported parameters.

B Visual Detection Parameters

Table 7: Main parameters for visual centre estimation.

Parameter	Value
Depth aggregation	
Retained frames: global / centred	9 / 7
Minimum valid observations per depth pixel	3
Global coarse detection	
Initial / additional sample-height interval	24–70 / 18–85 mm
Estimated diameter range	10–50 mm
Maximum aspect ratio	2.5
Minimum convex-hull circularity	0.45
Duplicate-distance threshold for additional candidates	40 mm
Required centre margin inside the current tray boundary	5 mm
Centred refinement	
Minimum camera clearance above the estimated sample top	280 mm
Accepted fitted radius	8–22 mm
Minimum angular coverage of inlier boundary samples	210°
Maximum inlier radial RMS residual	1.5 mm
Maximum centre-estimator disagreement	5 mm
Maximum refined-to-coarse centre displacement	50 mm

C Motion Parameters

Table 8: Motion parameters for sample handling.

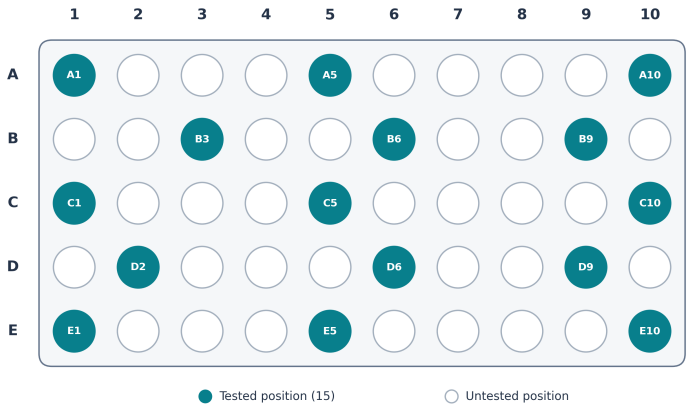
Parameter	Value
Horizontal transfer speed	0.30 m/s
Horizontal transfer acceleration	0.65 m/s^2
Vertical movement speed	0.08 m/s
Vertical movement acceleration	0.25 m/s^2
Workbench dwell time	2 s

D Communication Recovery Parameters

Table 9: Communication recovery and acquisition-validity settings.

Parameter	Value
Camera acquisition	
Acquisition attempts: overview / centred	2 / 3
Delay before reopening the pipeline	2 s
Centred-acquisition progress timeout	4 s
Maximum settling wait	5 s
Required settling interval	0.3 s
Maximum TCP linear speed during settling	2 mm/s
Maximum TCP angular speed during settling	0.02 rad/s
Maximum position change during capture	0.5 mm
Maximum orientation change during capture	0.15°
RTDE receive interface	
Configured receive frequency	125 Hz
Timestamp-advance checking window	0.60 s
Connection attempts	4
Waits before connection attempts	0, 0.5, 1.0, 1.5 s
RTDE control interface	
Connection attempts: normal / after RG2	4 / 6
Waits before normal connection attempts	0, 1, 2, 3 s
Waits before connection attempts after RG2	0, 2, 3, 5, 8, 12 s

E Supporting Material

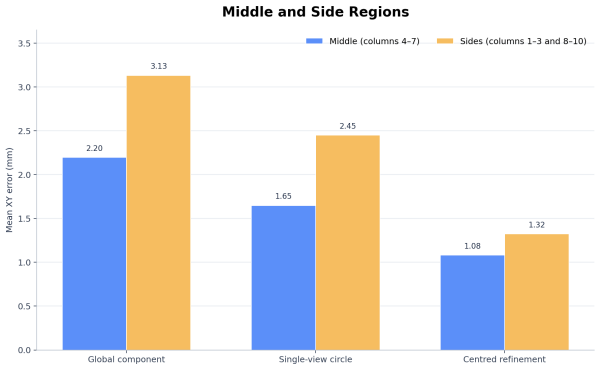


(a) Tray-hole numbering and tested positions.

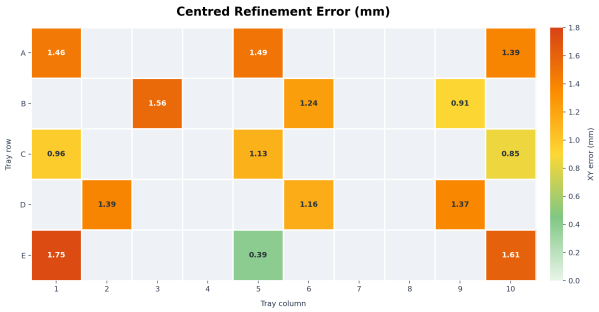


(b) Locating tool.

Figure 12: Supporting material for the evaluation in Section 4.3. (a) Tray-hole numbering, with the 15 tested positions highlighted; dimensions are not to scale. (b) Custom 3D-printed locating tool used to measure the reference centres.



(c) Mean errors in the middle and side regions.



(d) Position-level centred-refinement errors.

Figure 12: Supporting material for the evaluation in Section 4.3. (c) Mean errors in the middle region (columns 4–7) and side regions (columns 1–3 and 8–10), calculated from each method's valid positions. (d) Mean centred-refinement errors in millimetres; grey cells indicate untested positions.

F Workbench

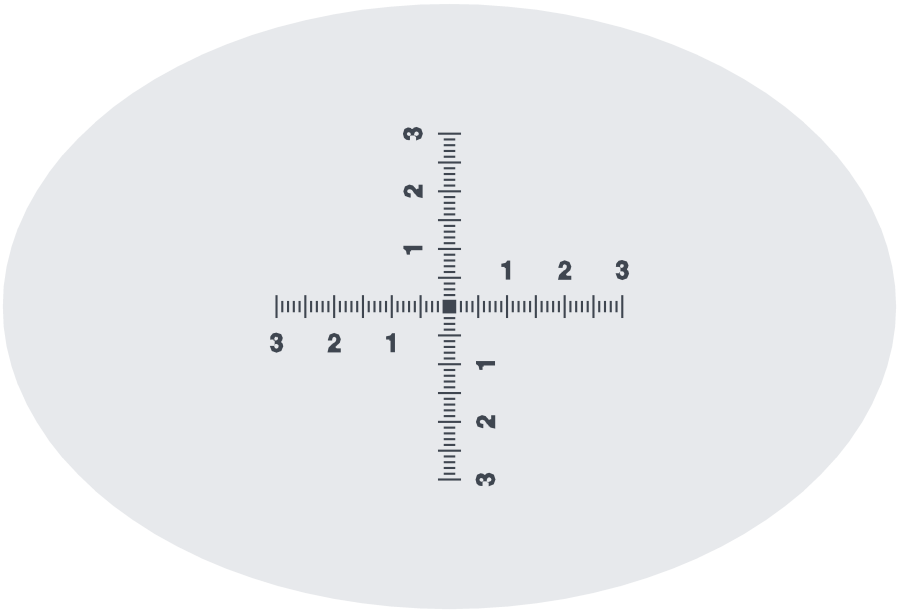


Figure 13: Top view of the experimental workbench.

G Code and Demonstration Videos

The source code, setup instructions, and demonstration videos are available in the project repository:

`https://github.com/lijiawei0113-beep/ur5e-sample-pick-return`